

Lattice-Reduction Attack as a Pipeline Stage: v8 Results and Why the Alternatives Failed

neuroMCTS project · Research meeting · July 2026

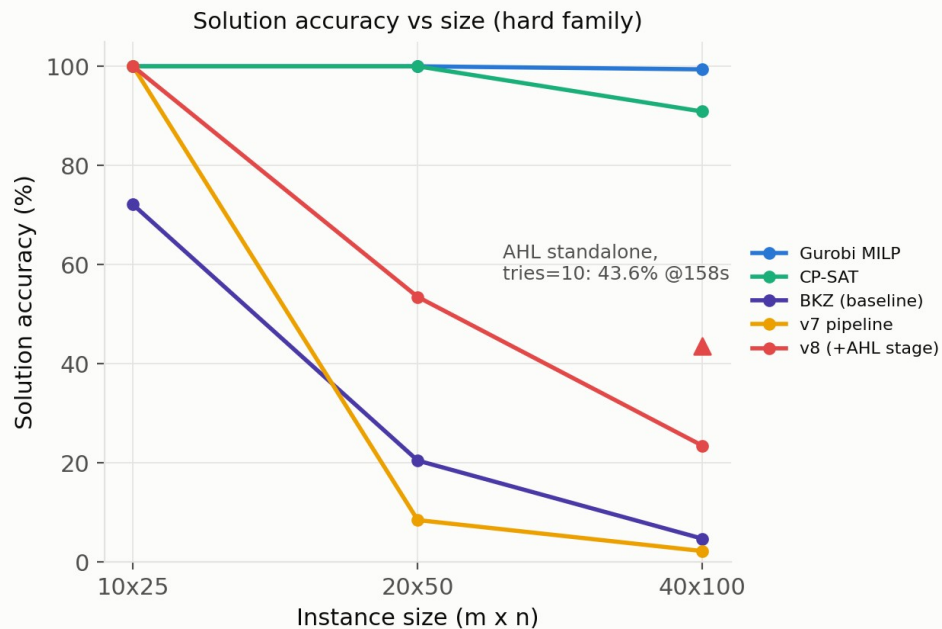
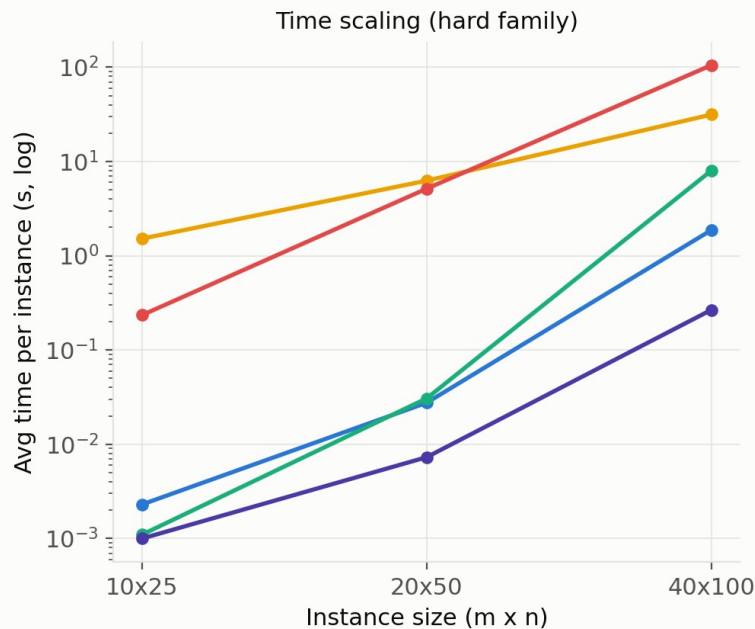
Three search technologies, one hard benchmark: only the AHL embedding works

Conflict learning (CBJ): 0/60 at 500k nodes — dense rows make conflict reasons vacuous

Kernel-lattice enumeration: 2/40 — box pruning too loose on dense kernels

AHL solution embedding + BKZ: 19/40 at 70 ms median — reduction reveals solutions directly

v8 = pipeline + AHL stage: strongest inexact method at every size



20x50: 9.6% -> 53.5% @5.2s | 40x100: 2.25% -> 23.5% (43.6% at full budget) | 10x25: 100% kept, 6.5x faster

Why it works where others fail — and where learning goes next

Solutions are short vectors ($2x-1 \mid -1 \mid 0$), norm $\sqrt{n+1}$: reduction finds them without search

Verified-only returns: zero false positives on infeasible inputs, zero false negatives kept

Learning roles ahead: try-scheduling (median 2 tries to hit; failures burn 10), hardness routing, fail-first proofs

Honest bound: exact solvers still ahead where they finish (Gurobi 99.4% @1.9s at 40x100)

References

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